

# FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

## Student Assessment Document

### FINAL EXPLANATION: CORE MATHEMATICS GRADE 10: STATISTICS I (3.1)

#### Student Assessment Document

|              |       |
|--------------|-------|
| Student Name | _____ |
| Class        | _____ |
| Date         | _____ |

### INSTRUCTIONS FOR STUDENTS

Final Explanation: Is Thika Road really well served? You have collected and studied data from the 8,000 m stretch of Thika Road near Ruaraka Hills Secondary School (the 20-gap sample). Answer each section in the space below it. Use numbers from our class data and always give the unit (metres, m). Write in full sentences. Where a question asks for a calculation, show your working. Remember: Ms Halima Abdi is not careless, so describe her claim as true or incomplete, not as wrong. Use the data table below for Section 2.

Grouped gap data, Thika Road, 20-gap sample:

| Class (m)     | Frequency |
|---------------|-----------|
| 0 to <200     | 5         |
| 200 to <400   | 9         |
| 400 to <600   | 0         |
| 600 to <800   | 2         |
| 800 to <1000  | 1         |
| 1000 to <1200 | 3         |
| Total         | 20        |

### 1. Checking the officer's claim

Ms Halima Abdi, the County Road Safety Officer, said: 'The average distance between crossing points is about 400 m, so the road is well served.' The stretch of Thika Road is 8,000 m long with 21 designated crossing points.

(a) How many gaps are there, and how can you check her figure of 400 m?

(b) Is her claim true? Is it complete? Explain using the ungrouped median and mode of the 20 gaps, which you worked out in class.

*Write your answer here:*

## 2. Reading the grouped table

Use the grouped table in the instructions.

- (a) Which is the modal class?
- (b) How many gaps are shorter than 400 m, and how many are 600 m or longer?
- (c) Calculate the estimated mean gap using class midpoints. Show your working and say why it is called an estimate.
- (d) The class 400 to <600 is empty, yet your estimated mean from (c) is above 400 m. What does this tell you about the mean?

*Write your answer here:*

### 3. Using graphs to see the shape

- (a) Describe what a histogram of the equal-width table would look like. Mention where the tallest bar is, the empty class and the bars on the right.
- (b) When classes 400 to <600, 600 to <800 and 800 to <1000 are merged into 400 to <1000, why must we use frequency density instead of frequency for bar height? Calculate the frequency density of this merged class (use class width in units of 100 m).
- (c) Mombasa Road (Embakasi) has a grouped mean of 380 m and Thika Road 440 m. Using their frequency polygons, explain why similar means can hide different stories.

*Write your answer here:*

#### 4. Where is the road really unsafe?

Use the pedestrian and near-miss data from the class project: the long stretch G9-G14 (six gaps from 600 m to 1,100 m) includes the school gate. The whole 20-gap sample had 182 pedestrians crossing at undesignated spots (Tuesday, 6:30-7:30 am), and Mzee Otieno's matatu drivers logged 12 near-misses.

(a) The long stretch had 140 of the 182 pedestrians. What percentage is this? What fraction of the road's 8,000 m does the long stretch (5,250 m) make up?

(b) Mzee Otieno's log has 9 of 12 near-misses in the long stretch. What percentage is this?

(c) Write a claim, evidence and reasoning paragraph that answers the driving question: How can we use data from our own road to show where pedestrians are really at risk, and why does one average not tell the whole story?

*Write your answer here:*

## 5. Recommendations and limits

- (a) What two recommendations would you give the school principal, the Parents Association, the area chief and the county roads representative? Support each with data.**
- (b) Give one reason why our findings should be treated carefully before we are fully certain.**
- (c) Answer the KICD question: How do we use statistics in day-to-day life? Use an example from this project.**

Write your answer here:

| RUBRIC                                               |                                                                                                                                                             |                                                                                                                     |                                                                                       |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Criterion                                            | Excellent (4)                                                                                                                                               | Proficient (3)                                                                                                      | Developing (1–2)                                                                      |
| Calculations and use of data                         | All calculations (mean, estimated mean, frequency density, percentages) are correct, show clear working and include units (m) and the 20-gap sample.        | Most calculations are correct with some working shown; occasional missing units or small slips.                     | Several calculation errors or no working shown; units often missing.                  |
| Understanding of mean, median, mode and grouped data | Explains clearly why the mean of 400 m is true but incomplete, using median 250 m, mode 200 m, and why grouped results are estimates.                       | States that the mean is true but incomplete and compares with median or mode, with limited explanation of grouping. | Only restates the mean or confuses mean, median and mode.                             |
| Interpreting graphs and tables                       | Accurately describes the histogram, frequency density and polygons, and links shape to where the long gaps are and why similar means hide different shapes. | Describes the main features of the graphs but gives limited comparison of Thika Road and Mombasa Road.              | Graph descriptions are incomplete or inaccurate; frequency density is not understood. |
| Claim, evidence and reasoning                        | Gives a clear claim supported by specific data (77%, 75%, G9-G14, 5,250 m) and reasoning that answers the driving question.                                 | Gives a claim with some correct evidence, but the reasoning is incomplete.                                          | Claim is unclear or lacks data evidence.                                              |
| Recommendations and                                  | Makes data-based recommendations,                                                                                                                           | Makes reasonable recommendations                                                                                    | Recommendations are not linked to data                                                |

|               |                                                                                          |                                                    |                 |
|---------------|------------------------------------------------------------------------------------------|----------------------------------------------------|-----------------|
| communication | states the one-morning limitation and links statistics to daily life with correct terms. | with some data support; limited mention of limits. | or are missing. |
|---------------|------------------------------------------------------------------------------------------|----------------------------------------------------|-----------------|

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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